

Unit 1.10 Problems in Real-World Contexts

Paper 1: Basic

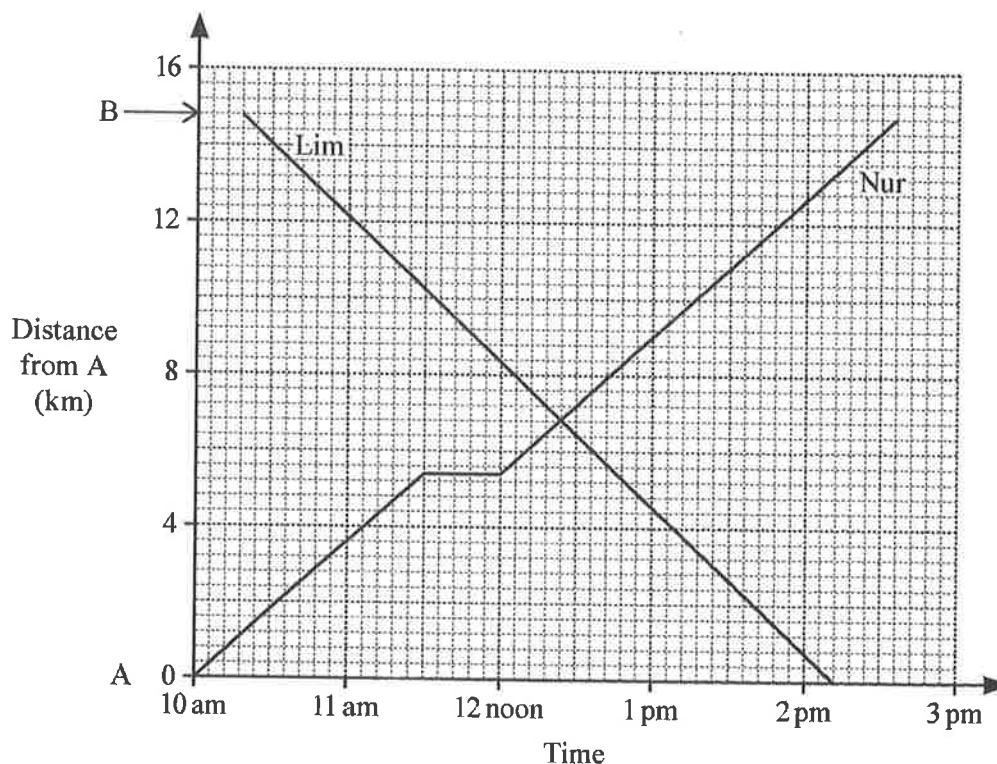
1. Sima invests \$2500 at a simple interest rate of 1.6% per year for 3 years.
Calculate the total value of her investment at the end of 3 years.

[2]
[N21/I/1]

2. Behnaz invests \$850 at a rate of $r\%$ per year compound interest.
At the end of 12 years the value of her investment is \$1120.
Calculate the value of r .

[3]
[N20/I/15]

3.



The distance-time graph shows the journeys for two people, Nur and Lim, between two villages, A and B.

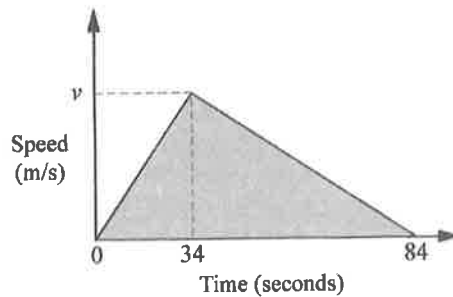
- (a) Find Nur's speed for the first hour of her journey.
(b) (i) Find the time when Nur and Lim pass each other.
(ii) How far are they from village B when they pass each other?
(c) Lim jogs back to village B at 8 km/h.
Change 8 km/h into metres per second.

[1]
[1]
[1]
[1]
[N19/I/17]

4. The cash price of a television is \$1475.
The hire-purchase price of the television is \$1651.
The hire-purchase price is a deposit plus 12 equal monthly payments of \$113.
Calculate the deposit as a percentage of the cash price.

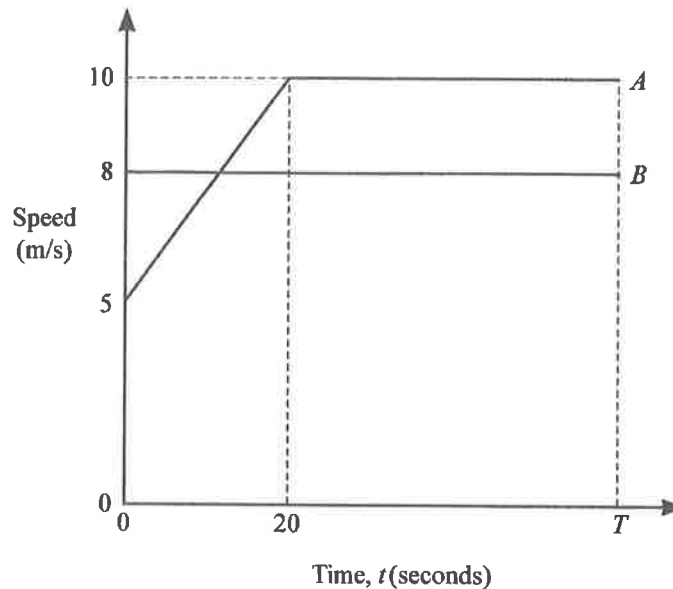
[3]
[N18/I/8]

5. The diagram shows the speed-time graph for a car's journey between two sets of traffic lights. The shaded area represents the distance travelled. The distance travelled is 693 m.



- (a) Calculate the greatest speed, v m/s, of the car. [1]
- (b) Calculate the acceleration for the first 34 seconds of the journey. [1]
- [N17/I/5]
6. Nur invested a sum of money in an account paying compound interest at 2.5% per year. After 4 years, the money had earned **total interest** of \$674.79. Calculate the sum of money Nur invested in the account. [3]
- [N17/I/13]
7. Frank bought an antique vase for \$345. One year later he sold it for a profit of 180% of the cost price. Calculate the selling price. [2]
- [N15/I/2]
8. Ravi invests \$28 700 in an account. The balance, \$ A , of the account after t years is given by the formula $A = 28\,700 \times 1.033^t$.
- (a) Calculate A when $t = 4$. [1]
- Give your answer correct to the nearest dollar. [2]
- (b) Find the percentage increase in the balance over the 4 years. [2]
- [N15/I/10]

9. The diagram shows speed-time graphs for two cyclists, A and B , travelling along the same road in the same direction. A accelerates for 20 seconds and then travels at a constant speed of 10 m/s. B travels at a constant speed of 8 m/s.



- (a) Calculate the acceleration of A for the first 20 seconds. [1]
- (b) The area beneath each speed-time graph represents the distance travelled by each cyclist.
- (i) Find an expression, in terms of T , for the distance travelled by B in the T seconds. [1]
- (ii) When $t = 0$, A is 90 m behind B .
When $t = T$, A passes B .
Find the value of T . [4]
- [N21/I/24]

10. In Singapore, Amy pays \$1.65 for one litre of diesel.

On a visit to Hawaii, she paid 4.48 US dollars for one gallon of diesel.

1 US dollar = 1.242 Singapore dollars.

1 gallon = 3.785 litres.

Is diesel cheaper in Singapore or Hawaii?

You must show your calculations.

[3]
[N16/I/12]

1. (a) Cheryl has some money to invest for 5 years.
 Account A pays 1% per year simple interest.
 Account B pays 1% per year compound interest.
 Explain why Account B is the better choice for Cheryl's investment. [1]
- (b) Marcus invests in an account that pays 1.8% per year simple interest.
 He leaves the money in the account for 6 years.
 At the end of 6 years there is \$1385 in the account.
 Calculate the amount of money Marcus invested. [3]
- (c) Tan invests in an account that pays 1.2% per year compound interest.
 He leaves the money in the account for 4 years.
 At the end of 4 years there is \$4719.92 in the account.
 Calculate the total amount of interest Tan earned over the 4 years. [3]
- (d) Helen hires a car during a business trip in France.
 She drives a total of 745 km.
 The car uses fuel at an average rate of 5.8 litres/100 km.
 She pays €134.50 for car hire and €1.52 per litre of fuel.
 She pays using her credit card and is charged a fee of 1.5% for the currency conversion.
 The exchange rate between Singapore dollars and euros is \$1 = €0.66.
 Calculate the total amount, including credit card fee, Helen is charged for car hire and fuel.
 Give your answer in Singapore dollars correct to the nearest cent. [4]
 [N22/II/2]

2. (a) The value of fuels and lubricants exported from Singapore in 2017 was \$90 182 million.
 Write \$90 182 million in standard form correct to 3 significant figures. [1]
- (b) The value of food exported from Singapore in 2012 was \$6230 million.
 The value of food exported from Singapore in 2017 was \$8007 million.
 Calculate the percentage increase in the value of food exported from 2012 to 2017. [2]
- (c) The value of machinery and equipment exported from Singapore in 2012 was $\$2.3 \times 10^{11}$.
 In 2017 this had increased by 9.3%.
 Calculate the value of machinery and equipment exported in 2017. [2]
- (d) From 2012 to 2017 the value of manufactured goods imported into Singapore decreased by 6.1%.
 In 2017 the value was $\$1.8 \times 10^{10}$.
 Calculate the value of manufactured goods imported in 2012. [2]
- (e) Jiao buys a camera in Japan costing 155 000 Yen.
 She pays GST at 7% on the cost of the camera when she returns to Singapore.
 The exchange rate between Singapore dollars and Japanese Yen is \$1 = 80.4 Yen.
 Calculate the amount, in dollars, Jiao spent on the camera including GST.
 Give your answer correct to the nearest dollar. [3]
 [N21/II/2]

3. (a) The price of a smartphone is \$620 in Singapore.
The price of the same smartphone in the UK is £389.
The exchange rate between Singapore dollars (\$) and UK pounds (£) is \$1 = £0.58.
Calculate how much cheaper the smartphone is in Singapore than in the UK. [2]

- (b) The price of a tablet is reduced by 6% in a sale.
The sale price of the tablet is \$785.
Calculate the price of the tablet before the sale.
Give your answer correct to the nearest dollar. [2]

- (c) The table shows information about telecommunications in Singapore.

Year	2010	2015
Mobile phone subscriptions	7.29×10^6	8.21×10^6
SMS messages sent	2.77×10^{10}	1.14×10^{10}
Broadband subscriptions	7.85×10^6	1.20×10^7

- (i) Calculate how many more mobile phone subscriptions there were in 2015 than in 2010.
Give your answer in standard form. [1]
- (ii) Calculate the percentage increase in the number of broadband subscriptions from 2010 to 2015. [2]
- (iii) The population of Singapore in 2015 was 5.54×10^6 .
Calculate the mean number of SMS messages sent per person per day in 2015. [3]
- [N19/II/3]

Paper 2: Intermediate

4. A small business makes jewellery.

Workers are paid a basic hourly rate with an additional payment for each item they make.

Basic rate	\$9.80 per hour
Earrings	\$2.50 per pair
Necklace	\$1.65 each
Bracelet	\$1.45 each
Brooch	\$0.85 each

These are the employment guidelines provided for the workers.

Total of 40 hours per week over 5 days
18 days annual holiday paid at basic hourly rate
Expected annual income at least \$48 000

- (a) Abid makes 7 pairs of earrings in one hour.
Calculate the time he takes to make one pair of earrings.
Give your answer correct to the nearest 10 seconds. [1]
- (b) Mei makes bracelets.
One day she works for 9 hours and earns a total of \$231.75.
Show that Mei made an average of 11 bracelets per hour. [2]
- (c) In one day, a total of 132 necklaces are made.
Chen and Zhu make these necklaces.
Zhu takes 80 seconds less than Chen does to make each necklace.
They each work for 8 hours a day.
Can Chen and Zhu each expect to earn the advertised minimum annual income?
Justify your decision and show your method clearly. [8]

[N22/II/10]

5. Ying is going to study in London for a year.

She needs to work out how much money, in Singapore dollars, she will need to cover her living expenses.

She finds the information below to help her work out her expenses.

The costs are all given in pounds (£).

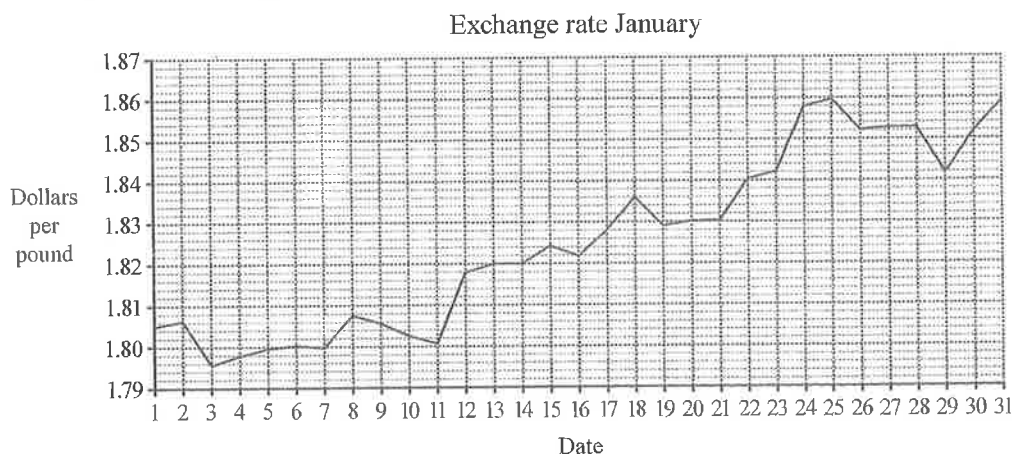
Estimated student living costs in London

Accommodation	£217 per week
Student travel pass	£56.90 per month

Item	Cost per week
Food and household goods	£60
Personal items	£57
Leisure items	£25
Books and equipment	£15

- (a) The exchange rate between pounds and Singapore dollars changes each day.

The graph shows the daily exchange rate for January.



- (i) Write down the exchange rate between pounds and Singapore dollars on 15 January.

Give your answer correct to 3 decimal places.

[1]

- (ii) Ying wants to convert her weekly accommodation cost to dollars.

Use the graph above to work out the difference between the greatest and least possible weekly accommodation costs in dollars.

Give your answer correct to the nearest cent.

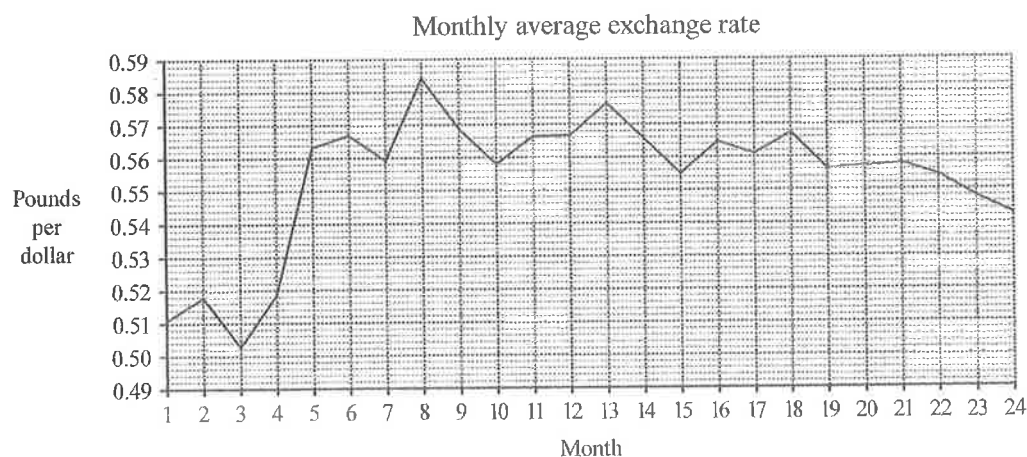
[3]

- (b) Ying's parents will give her an allowance, in dollars, to cover her total costs including accommodation and travel.

She must tell them how much money she will need for the year.

Her parents add an extra 15% to this amount to cover any extra costs.

Ying finds this information about exchange rates over the past two years.



Suggest a suitable amount in dollars for Ying's parents to give her for the year.

Justify any decisions that you make and show your calculations clearly.

[6]

[N20/II/10]

6. Leila lives in Kuala Lumpur and drives a small family car.

The tables below give information that Leila can use to work out her driving costs.

Type of driving	Amount of fuel used (litres/100 km)			
	Type of car			
	Supermini	Small family	Large family	SUV
City	5.5	6.3	7.8	7.6
Out of city	3.9	4.2	4.6	5.0
Combined	4.5	5.0	5.8	6.0

Speed limits	
City roads	60 km/h
Out of city	
Federal roads	90 km/h
Expressway	110 km/h

Fuel prices per litre	
Regular	\$2.07
Premium	\$2.11
Diesel	\$1.45
5% discount with loyalty card	

- (a) One week, Leila drives a total distance of 92 km in the city.

Leila estimates that she uses 5.8 litres of fuel.

Show that Leila is correct.

[1]

- (b) Leila drives for 45 minutes on the expressway at an average speed of 85 km/h.

Calculate an estimate of the amount of fuel she uses.

[2]

- (c) Leila and Hamid go on a journey together in Leila's car.

They drive from the city of Kuala Lumpur to a town outside of Kuala Lumpur.

Leila drives the first stage of the journey at an average speed of 60 km/h.

The second stage of the journey is 25 km shorter than the first stage.

Leila drives the second stage at an average speed of 75 km/h.

The journey takes a total of 3 hours 15 minutes.

Hamid offers to pay half of the cost of the fuel used for the journey.

Leila's car uses regular fuel and she has a loyalty card.

Suggest a suitable amount for Leila to ask Hamid to pay.

Justify the decision you make and show your calculations clearly.

[7]

[N19/II/10]

7. (a) In 2016, Ken earned a total of \$44 000.

In 2017, he earned \$3900 each month.

Calculate the percentage increase in his earnings from 2016 to 2017.

[2]

- (b) A bank offers a savings account with a rate of compound interest of 2.35% per year.

Ken invests \$6500 in this account.

He leaves the money in the account for 6 years.

Calculate the total amount of interest he has earned after 6 years.

Give your answer correct to the nearest cent.

[3]

- (c) The exchange rate between Singapore dollars (\$) and euros (€) is $\$1 = \text{€}0.66$.
 The exchange rate between pounds (£) and Singapore dollars is $\text{£}1 = \$1.97$.
 Ken is planning a trip to Europe.
 He finds these hotel prices on a website.

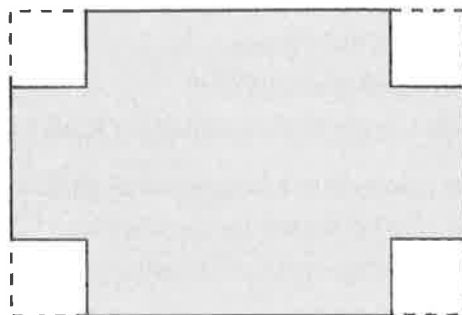
London Hotel £145 per night
 Paris Hotel €145 per night

- (i) By comparing the exchange rates, explain how you can tell that the hotel in London costs more per night than the hotel in Paris. [1]
- (ii) Ken books 3 nights in the hotel in London and 4 nights in the hotel in Paris.
 He pays using his credit card.
 The credit card company converts the prices to Singapore dollars.
 Ken is charged a fee of 1.8% for the currency conversion.
 Calculate the total amount Ken pays for the two hotels, including the credit card fee.
 Give your answer correct to the nearest dollar. [4]

[N18/II/2]

8. Answer the whole of this question in the graph paper booklet.

Huan runs a business selling sweets.
 She is designing a container to hold the sweets.
 She uses rectangular pieces of cardboard measuring 20 cm by 30 cm.
 The mass of 1 square metre of the cardboard is 210 grams.



She cuts a square from each corner of a piece of cardboard.
 She then folds the cardboard to make an open box.

By changing the size of the square she cuts out, Huan can change the volume of the box.

- (a) Work out the mass of the box made when she cuts a square of side 5 cm from each corner. [2]
- (b) Huan cuts a square of side x from each corner of the cardboard.
- (i) Explain why the volume of the box she makes is given by $x(20 - 2x)(30 - 2x)$. [1]
- (ii) Explain why $0 < x < 10$. [1]
- (c) Huan wants to make a box with a volume of at least 900 cm^3 .
 She wants the mass of the box to be as small as possible.
 By drawing a suitable graph, work out the mass of the box she will make. [7]

[N18/II/10]

9. (a) Noor works in a shop.
She is paid \$10.50 for each hour she works.
She is also paid a bonus of 6% of the value of the goods she sells.
One week she works for 32 hours and sells goods to the value of \$1350.
Calculate her total earnings for the week. [2]
- (b) (i) The cash price of a TV is \$3499.
Sue buys this TV on hire purchase.
She pays a deposit of one fifth of the cash price.
She then makes 24 monthly payments of \$130.
Find the total amount that Sue pays for the TV. [2]
- (ii) A salesman sells a set of TV speakers to Sue for \$342.
He makes a loss of 5% on the price he paid for the speakers.
Calculate the price the salesman paid for the speakers. [2]
- (c) (i) John visits his brother in the UK.
To pay for the flight, he borrows \$1500 for 3 years at compound interest of 7.5% per year.
Calculate how much interest John will pay on the loan.
Give your answer correct to the nearest cent. [3]
- (ii) John spends £185 on his credit card when he is in the UK.
He pays a credit card fee of 1.5% of this amount.
The credit card company uses an exchange rate between Singapore dollars (\$) and pounds (£) of \$1 = £0.52.
Calculate the total cost in Singapore dollars that John has to pay the credit card company.
Give your answer correct to the nearest cent. [3]

[N15/II/3]

10. Nurul is planning a weekly exercise routine.

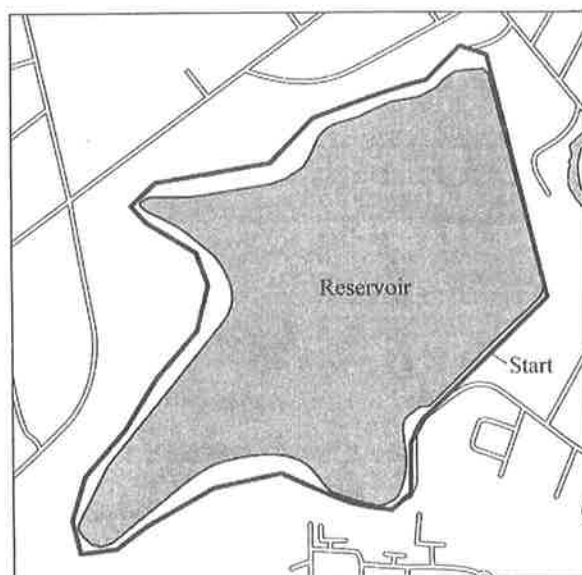
She has read the following health advice:

Time: For optimal health benefits, aim to accumulate 150 minutes of moderate-intensity or 75 minutes of vigorous-intensity aerobic activity each week. Some individuals may need to accumulate 250 minutes a week to prevent weight gain.

1 minute of vigorous-intensity aerobic activity = 2 minutes of moderate-intensity aerobic activity, e.g. 10 minutes of jogging = 20 minutes of brisk walking

She plans to go for three brisk walks each week.

The map shows her planned route around her local reservoir.



Scale: 1:20 000

Key

— Nurul's route

- (a) Estimate the length of Nurul's route. [2]
- (b) Nurul aims to walk at 6.5 km/h.

Does Nurul meet the weekly time target recommended in the health advice if she walks at this speed?

Show how you decide. [2]

- (c) To use more calories during the exercise, Nurul decides to jog the same route three times each week instead of walking.

Nurul weighs 70 kg and she estimates that her jogging speed will be 9.5 km/h.

She finds the following information on the internet.

Approximate calories used during 30 minutes of exercise

	Body weight			
	60 kg	70 kg	80 kg	90 kg
Walking 5 km/h	105	120	135	150
Walking 6.5 km/h	130	150	170	190
Jogging 8 km/h	240	280	315	350
Jogging 9.5 km/h	300	345	390	435

Nurul thinks that she will meet the target for time and that she will use more than double the amount of calories.

Is Nurul correct?

Justify your decision with calculations.

[6]

[N17/II/10]

11. Peter produces the newsletter for a sports club.

Each newsletter is printed in black on both sides of four sheets of A4 paper.

Peter posts the newsletter to subscribers 6 times each year.

Information that Peter needs is on the next page.

Peter uses a C5 envelope to post the newsletter. He finds that a C5 envelope containing 4 sheets of A4 paper weighs approximately 25 g.

(a) How much will it cost to post one copy of the newsletter? [1]

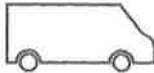
He estimates that there will be 150 subscribers to the newsletter.

(b) How many toner cartridges will Peter need to print the newsletters for one year? [2]

Peter needs to decide how much he should charge subscribers for the newsletter.

He must make sure that he charges enough money to cover all of his costs.

(c) Suggest a sensible amount for him to charge for a one-year subscription to the newsletter. [7]


LOCAL MAIL RATES (inclusive of 7% GST)

For Letters, Postcards, Printed Papers and Packets/Packages		
Weight Step Up To	Standard Mail	Non-Standard Mail
Regular		
20 g	\$0.30	\$0.60
40 g	\$0.37	
Large		
100 g	\$0.60	\$0.90
250 g	\$0.90	\$1.15
500 g	\$1.15	\$1.70
1 kg	\$2.55	
2 kg	\$3.35	

Standard regular mail
(C5, C6 and DL size envelope)

Standard large mail
(up to C4 size envelope)

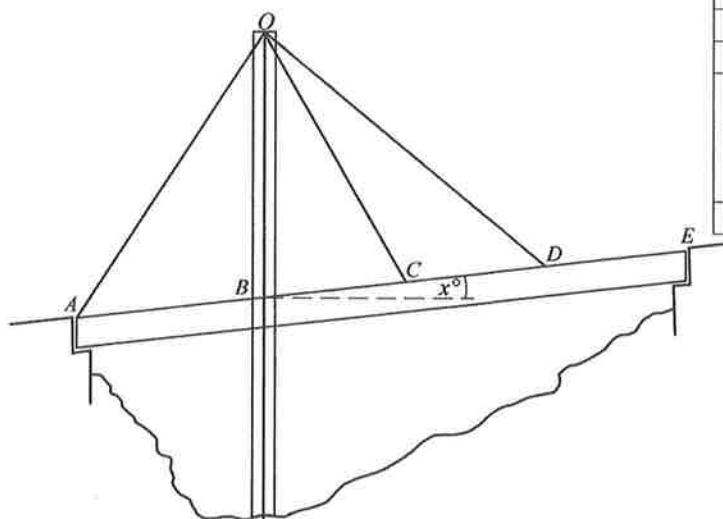
Stationery supplies (costs excluding 7% GST)		
Item	Description	Unit cost
A4 paper	White printer paper	
	Pack of 500 sheets	\$3.90
	Bulk price (10+ packs)	\$3.50
	Bulk price (30+ packs)	\$3.30
Toner cartridges	Black printing	\$147.00
	Average yield 1300 pages	
	Colour printing	\$151.00
	Cyan/yellow/magenta cartridges	
C5 envelopes	Average yield 1000 pages	
	White self-seal envelopes	
	Pack of 10	\$1.50
	Pack of 500	\$34.00

[N16/II/10]

Unit 2.8 Problems in Real-World Contexts

Paper 2: Intermediate

1.



Key dimensions		
Part		Length (m)
Mast	OB	35.0
	AB	30.0
Deck	BC	20.0
	CD	20.0
	DE	20.0
Cable	OA	48.3

The diagram represents a cable-stayed footbridge.

The deck $ABCDE$ is supported by three cables, OA , OC and OD .

The cables are suspended from the vertical mast OB .

The deck is at an angle of x° to the horizontal.

Cable OA has diameter 64 mm.

Cable OC and OD both have diameter 44 mm.

Information about cables used in bridge building is shown in the table below.

Product code	Diameter (mm)	Design load (kN)	Unit mass (kg/m)
FLC 32	32	615	5.7
FLC 36	36	779	7.2
FLC 40	40	973	9.0
FLC 44	44	1179	10.8
FLC 48	48	1403	12.9
FLC 52	52	1667	15.3
FLC 56	56	1933	17.8
FLC 60	60	2218	20.4
FLC 64	64	2524	23.2
FLC 68	68	2848	26.2

- (a) The gradient of the bridge deck is such that the ratio vertical distance : horizontal distance is 1 : 10.
Use the ratio to show that $x = 5.7$, correct to 1 decimal place. [1]
- (b) Show that $OC = 38.5$ m, correct to 1 decimal place. [3]
- (c) The bridge engineer has specified a maximum of 2.1 tonnes for the total mass of the three cables.
1 tonne is equal to 1000 kg.
Do the cables meet the engineer's specification?
Justify your decision. [6]

[N21/II/10]

